

AN ENSEMBLE LEARNING APPROACH FOR IMPROVED SLEEP DISORDER PREDICTION

SYNOPSIS

Submitted in partial fulfillment for the award of the degree of

**Master of Technology
in
Artificial Intelligence and Data Science**

Submitted by

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Under the Guidance of

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**SCHOOL OF ENGINEERING AND TECHNOLOGY
SRI PADMAVATI MAHILA VISVAVIDYALAYAM
(Women's University)**

**Tirupati
May 2026**



Department of Computer Science and Engineering

DECLARATION BY THE CANDIDATE

I hereby declare that the Report entitled “**AN ENSEMBLE LEARNING APPROACH FOR IMPROVED SLEEP DISORDER PREDICTION**” submitted by **P. Bhargavi (24MT8202)** to the Department of Computer Science and Engineering, School of Engineering and Technology, Sri Padmavati Mahila Visvavidyalayam, Tirupati in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Artificial Intelligence and Data Science** is a record of bonafide work carried out by me under the supervision of **Ms. G. Sreehitha, Assistant Professor** and **Ms. G. Radhika, Assistant Professor**. I further declare that the work reported has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this university or any other institute or university.

Place:

Signature of the Candidate

Date:



Department of Computer Science and Engineering

BONAFIDE CERTIFICATE

This is to certify that the report entitled **“AN ENSEMBLE LEARNING APPROACH FOR IMPROVED SLEEP DISORDER PREDICTION”** submitted by **P. Bhargavi (24MT8202)** to the Department of Computer Science and Engineering, School of Engineering and Technology, Sri Padmavati Mahila Visvavidyalayam, Tirupati in partial fulfillment of the requirement for the award of the degree of **M. Tech in Artificial Intelligence and Data Science** is a record of bonafide work carried out by her under our guidance. The work fulfills the requirements as per the regulations of this University and in my opinion, meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

Sleep disorders can harm both physical and mental health, but traditional diagnostic methods like Polysomnography (PSG) are costly, uncomfortable, and not easily available. The system collects user inputs such as age, occupation, sleep duration, caffeine intake, blood pressure, heart rate, BMI category, snoring frequency, working hours and health-related data to predict possible sleep disorders. This study aims to develop a more accessible and accurate method for detecting sleep disorders using machine learning techniques. Instead of relying on single models such as K-Nearest Neighbours (KNN), Support Vector Machine (SVM), and Random Forest. We combine multiple models using ensemble techniques like Stacking or Voting Classifiers. The system shows better performance in identifying sleep disorders. The model is integrated into a mobile application to provide a fast, cost-effective, and user-friendly tool for detecting sleep disorders. This approach offers a practical alternative to traditional tests and may help improve patient outcomes.

Keywords: Sleep Disorders, Machine Learning, Ensemble Learning, Voting Classifiers, Mobile Application.

1. Introduction

Sleep is essential for maintaining good physical health, mental stability, and daily performance, but traditional diagnostic methods like Polysomnography (PSG) are costly, uncomfortable, and not easily available. This makes diagnosis difficult, especially for people in rural or low-resource areas. In many cases, individuals are unaware of their sleep-related issues until they become severe, leading to long-term health complications such as stress, cardiovascular problems, and reduced work efficiency. The lack of awareness and limited access to diagnostic facilities further increases the risk of untreated sleep disorders. To address these challenges, this project uses machine learning to provide a simpler and more accessible method for identifying sleep disorders. By analyzing basic health and lifestyle information, machine learning models can quickly predict whether a person is likely to have a disorder. The mobile application (APK) is developed by converting the web application into a portable format, ensuring accessibility across different devices. This integration of web and mobile platforms enhances user convenience and ensures that the system can be used by a wider audience. Overall, the proposed solution provides a practical, efficient, and user-friendly approach for early detection and monitoring of sleep disorders.

2. Thesis Statement

Sleep disorders are increasingly affecting people's health and daily life, but early diagnosis remains a major challenge. Traditional diagnostic methods such as Polysomnography (PSG) are expensive, time-consuming, and require specialized medical facilities, making them inaccessible to many individuals, especially in rural and low-resource areas. There is no simple and easily available system that allows users to assess their sleep health using basic personal and lifestyle information. As a result, many people remain undiagnosed or delay seeking medical help until the condition becomes severe. Therefore, there is a need to develop a cost-effective, accessible, and user-friendly system that can predict sleep disorders at an early stage using machine learning techniques. The system should be easily accessible through a web link and mobile application, enabling users to get quick predictions anytime and anywhere.

3. Objective of the Thesis

To develop an efficient and user-friendly system for predicting sleep disorders using machine learning techniques. To provide an accessible and cost-effective solution for early

detection of sleep-related issues. To analyze user input data such as lifestyle habits, health parameters, and sleep patterns for accurate prediction. To assist individuals in identifying potential sleep disorders at an early stage and encourage timely medical consultation.

4.Literature Survey

Muhammad Mostafa Monowar, S. M. Nuruzzaman Nobel, Maharin Afroj, “Advanced Sleep Disorder Detection Using Multi-Layered Ensemble Learning and Advanced Data Balancing Techniques”, 2025 [1]. The existing system focuses on detecting sleep disorders using single machine learning models; however, their accuracy and stability often vary, making them less reliable for consistent predictions. Although the proposed multi-layered ensemble approach improves performance compared to individual models, it introduces several challenges. The model is computationally intensive, requiring high processing power and longer training time, which makes it less suitable for real-time or resource-constrained environments. Additionally, there is a higher risk of overfitting due to the complexity of the ensemble techniques, especially when trained on limited datasets. Another significant limitation is the lack of validation using diverse external datasets, which affects the model’s ability to generalize well in real-world scenarios. Furthermore, the system may face difficulties in scalability, integration with practical healthcare systems, and handling noisy or incomplete user data, thereby limiting its widespread deployment and practical usability.

MD Kamal Rasel, Taanzina Afroz Rimi, Mahmuda Rawnak Jahan, Shayla Sharmin, “Improving Sleep Disorder Diagnosis Through Optimized Machine Learning Approaches” [2]. The primary limitation of this study is the use of a very small and limited dataset, which significantly reduces the reliability and real-world accuracy of the developed models. Due to insufficient data, the model may not capture diverse sleep patterns and health conditions, leading to poor generalization when applied to a larger population. The study also lacks validation with external or real-time datasets, which further limits its practical applicability. Moreover, the performance of the optimized machine learning techniques may not remain consistent across different demographic groups. The system may also struggle with handling noisy or incomplete data, and its scalability for large-scale deployment remains uncertain, thereby restricting its effectiveness in real-world healthcare applications.

C. Tran, Y. Wijesuriya, R. Thuraisingham, A. Craig, & H. Nguyen, "Deep Learning for Classification of Sleep Stages," 2019 [3]. This study focuses on the application of deep

learning techniques for the classification of sleep stages using EEG data. The authors highlight the ability of deep learning models to effectively handle complex and high-dimensional sleep data, resulting in high classification accuracy.

E. Alickovic & A. Subasi, "Ensemble SVM Method for Automatic Sleep Stage Classification," IEEE Transactions on Instrumentation and Measurement, 2018 [4]. This paper discusses the use of ensemble Support Vector Machine (SVM) methods for automatic sleep stage classification. The authors address the challenges of accurately classifying sleep stages and demonstrate the effectiveness of ensemble techniques in improving overall classification accuracy compared to single SVM models. Their approach combines multiple SVM classifiers to enhance robustness and reduce misclassification errors. The study highlights that ensemble learning can better handle variability in physiological signals such as EEG, leading to more stable and reliable predictions. Additionally, the method improves generalization performance and reduces the impact of noise in the data. However, the approach increases computational complexity and requires careful selection of parameters and model combinations. The study also indicates that the performance depends on the quality of input data and may require preprocessing steps for optimal results, which can add to system complexity.

M. J. Sun, Z. F. Wu, & X. B. Lu, "Sleep Apnea Detection Based on Time and Frequency Domain Analysis of ECG and SpO2 Signals," 2020 [5]. The authors propose a method for detecting sleep apnea using time and frequency domain analysis of ECG and SpO2 signals. By applying machine learning algorithms to these physiological signals, they achieve high accuracy in identifying sleep apnea events. This research highlights the importance of combining multiple physiological data sources to improve detection performance. The use of both time-domain and frequency-domain features allows the model to capture detailed patterns and variations in heart rate and oxygen saturation levels, leading to more precise predictions. Additionally, advanced signal processing techniques help in reducing noise and improving the quality of input data. The study also demonstrates that integrating ECG and SpO2 signals enhances the reliability of the system compared to using a single data source.

T. Radha, V. S. Kumar, & S. Pradeep, "Classification of Sleep Disorders Using Machine Learning Algorithms," International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 2019 [6]. This study explores the use of various machine learning algorithms for the classification of sleep disorders based on polysomnography (PSG) data. The authors compare the performance of different models,

including decision trees, random forests, and neural networks, highlighting the advantages and limitations of each approach. Their results demonstrate the strong potential of machine learning techniques in improving the accuracy and efficiency of sleep disorder diagnosis. Additionally, the study shows that ensemble methods like random forests provide better performance due to their ability to reduce overfitting and handle complex data patterns. Neural networks, on the other hand, are effective in capturing non-linear relationships in PSG data, leading to improved classification results. The research also emphasizes the importance of feature selection and data preprocessing in enhancing model performance. However, the study indicates that the accuracy of these models depends heavily on the quality and size of the dataset. Furthermore, the use of PSG data requires specialized equipment and clinical settings, which may limit large-scale or real-time applications. The computational complexity of advanced models and the need for expert interpretation can also pose challenges for practical deployment.

K. Vuppalapati, V. Guddeti, & P. Prasad, "Sleep Disorder Classification Using EEG Signal Analysis and Machine Learning," 2021 [7]. The authors present a comprehensive approach to sleep disorder classification using EEG signal analysis and machine learning techniques. By extracting relevant features from EEG signals and applying classifiers such as SVM and deep neural networks, they achieve high classification accuracy. This research emphasizes the significance of feature extraction and model selection in the context of sleep disorder classification.

5. Proposed System

5.1 Introduction

The proposed system aims to provide a more accurate, accessible, and user-friendly solution for detecting sleep disorders using machine learning. Unlike traditional Polysomnography (PSG), which is expensive and inconvenient, this system uses ensemble learning techniques such as Stacking or Voting Classifiers to combine multiple predictive models, enhancing accuracy, stability, and interpretability. The system processes data as inputs from users, enabling it to detect various sleep disorders efficiently. It is integrated into a mobile application, allowing users to input personal details, track their sleep patterns, and receive instant reports on potential sleep issues. Additionally, the system supports physician interaction, enabling doctors to review the results and recommend treatments or lifestyle adjustments if necessary. By combining machine learning, wearable-compatible data collection, and mobile accessibility, the proposed system provides a fast, cost-effective, and practical alternative to traditional sleep diagnostics, improving early detection and patient care.

5.2 Architecture

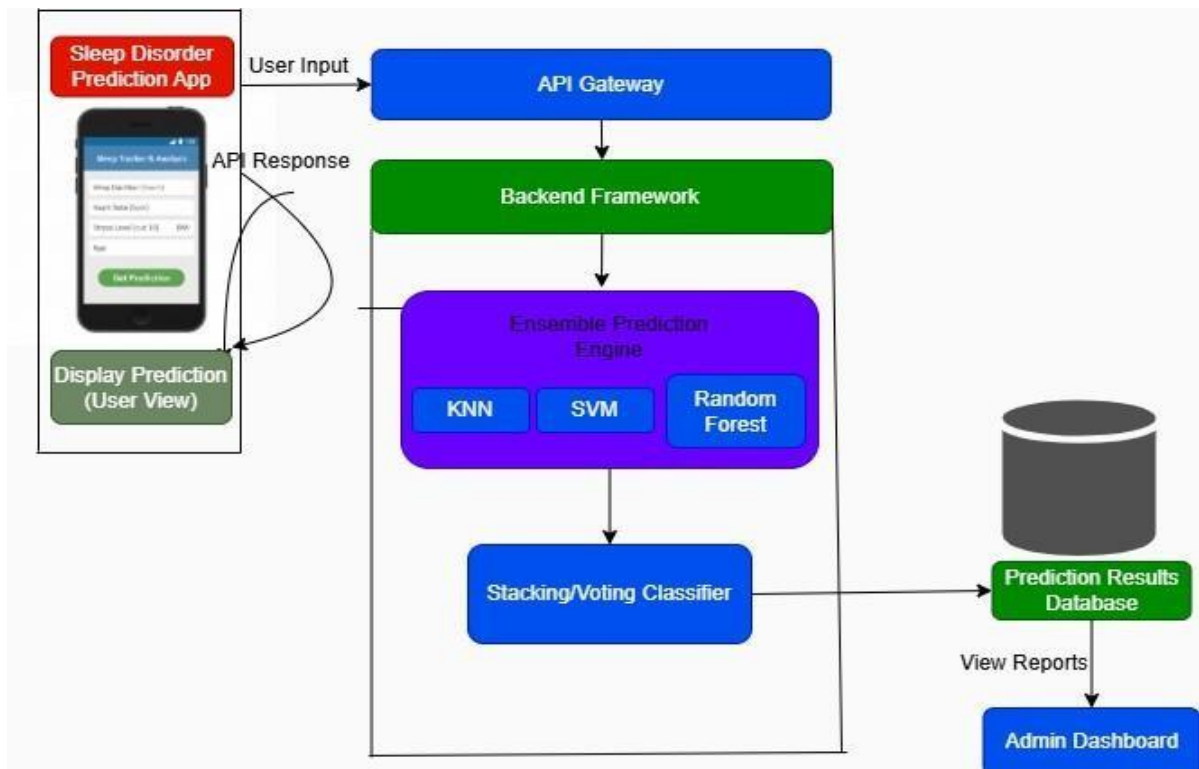


Fig 1: Architecture

5.3 Modules Description

The project is carried out based on the following modules listed:

1. Client-Side (User View)

The Sleep Disorder Prediction App serves as the frontend interface through which users interact with the system. It allows users to input their health-related information such as sleep duration, heart rate, stress level, BMI, and age in a simple and user-friendly manner. The application is designed with proper input validation to ensure that the data entered by users is accurate and in the correct format. Additionally, the interface can be integrated with wearable devices such as smartwatches and fitness bands to automatically collect real-time health data, thereby improving the accuracy and convenience of the system. Once the data is processed by the backend, the application displays the prediction results directly to the user. These results may indicate conditions such as Healthy, Insomnia, or Sleep Apnea. Along with the prediction, the system can also provide visual representations like graphs and trends to help users better understand their sleep patterns. Furthermore, personalized recommendations and suggestions can be provided to improve sleep quality, and users can track their past reports for continuous monitoring of their health.

2. Communication & Processing Layer

The API Gateway acts as the entry point of the system, handling all incoming requests from the client side application. It is responsible for receiving user input, validating the request format, and securely routing it to the backend framework for further processing. In addition to routing, the API Gateway also manages important functions such as authentication, authorization, request logging, and load balancing, ensuring that the system remains secure, reliable, and scalable. The Backend Framework forms the core processing unit of the system and can be implemented using technologies such as Node.js, Express, or Django. It is responsible for handling the overall business logic of the application. After receiving data from the API Gateway, the backend performs preprocessing steps such as data cleaning, normalization, and feature selection to prepare the data for machine learning models. It also manages error handling and coordinates communication between different system components, including the database and the prediction engine.

3. Ensemble Prediction Engine (Core Intelligence Layer)

The Ensemble Prediction Engine is the central component responsible for generating accurate predictions. It consists of multiple base machine learning models, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest. Each of these models independently analyzes the input data and produces its own prediction based on different learning techniques. KNN works by identifying similar data points and predicting based on majority similarity, SVM separates data using optimal decision boundaries, and Random Forest uses multiple decision trees to improve accuracy and reduce overfitting. To enhance prediction performance, the system uses ensemble techniques such as Voting or Stacking. In the voting approach, the final prediction is determined based on the majority decision of all models. In stacking, a meta-model is used to combine the outputs of base models for improved accuracy. The final output generated by the ensemble engine may classify the user's condition as Healthy, Insomnia, or Sleep Apnea. Additionally, the system can provide confidence scores and explanations for predictions, making the results more interpretable and reliable.

4. Data Storage & Management Layer

The Prediction Results Database serves as a centralized storage system where all user inputs, prediction results, and historical data are securely stored. This ensures that data can be accessed later for analysis, tracking, or reporting purposes. The database is designed with security and scalability in mind, incorporating features such as data encryption, backup mechanisms, and cloud-based storage support to handle large volumes of data efficiently. The Admin Dashboard is a dedicated interface used by healthcare professionals or system administrators to monitor and analyze patient data. It retrieves information from the database and provides functionalities such as viewing individual reports, analyzing trends, and exporting data for further use. Advanced features such as data visualization through charts and graphs, patient monitoring systems, and decision-support tools can assist doctors in making informed decisions and providing better treatment recommendations.

6.Implementation

The implementation phase is a crucial stage of the project where the designed system is transformed into a working model. This phase ensures that the proposed system operates effectively and meets user requirements. It also helps in building user confidence by demonstrating accurate and reliable results.

6.1 Output Screenshots

6.1.1 Home Page: This is home page of my website.

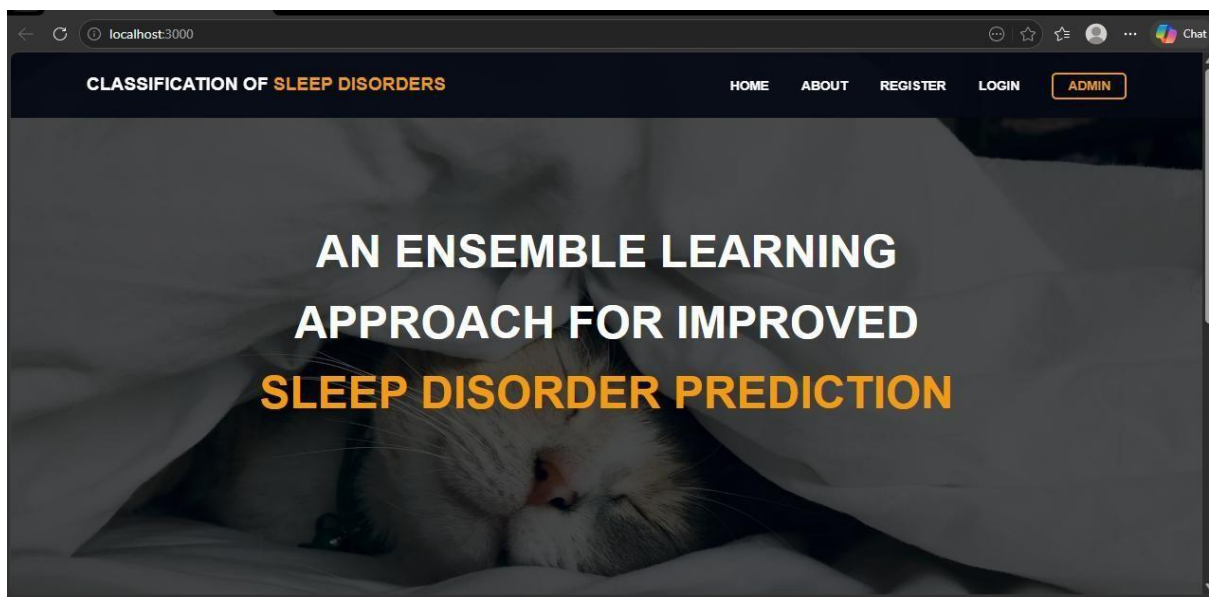


Fig2: Home Page

6.1.2 About Page: This is about section which contains information about my project.

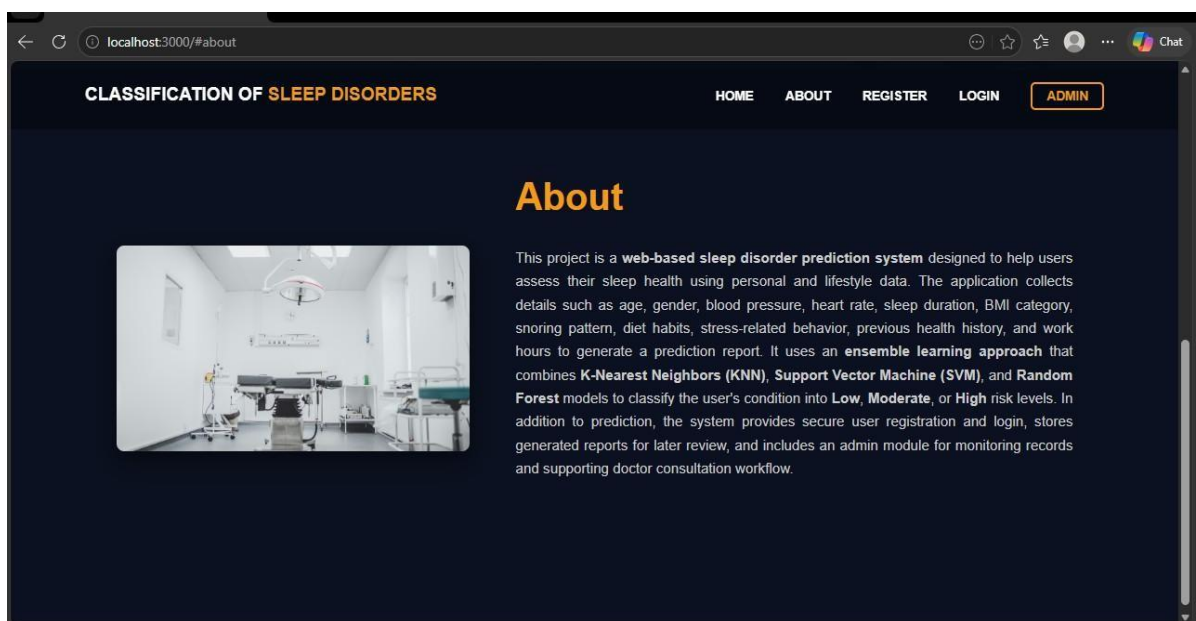
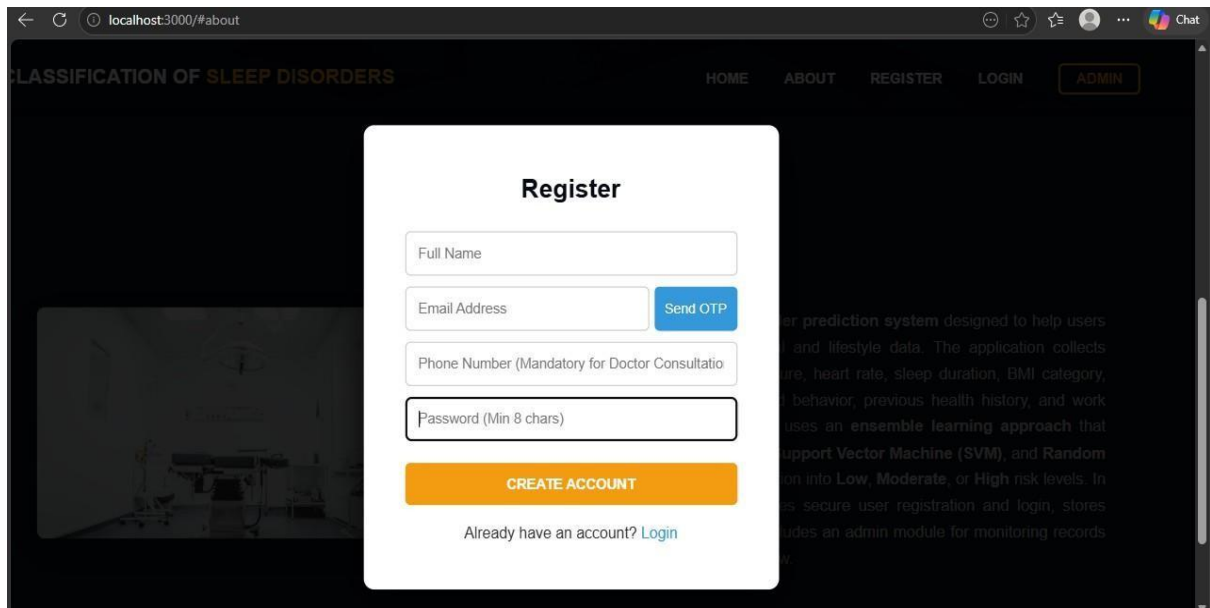


Fig 3: About Page

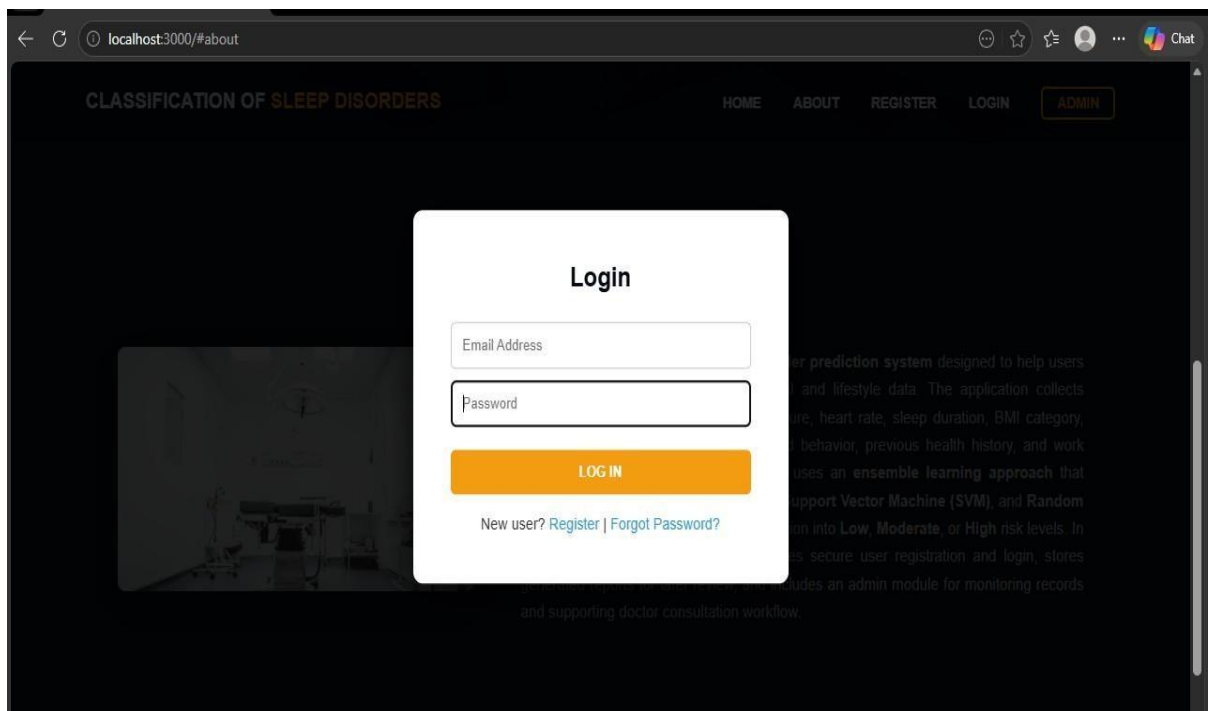
6.1.3 Registration Page: This is Registration page. In here, user can register with their credentials.



The screenshot shows a web browser at localhost:3000/#about. The page title is "CLASSIFICATION OF SLEEP DISORDERS". The navigation bar includes links for HOME, ABOUT, REGISTER, LOGIN, and an ADMIN button. A central white modal titled "Register" contains the following fields: "Full Name", "Email Address" (with a "Send OTP" button), "Phone Number (Mandatory for Doctor Consultation)", and "Password (Min 8 chars)". Below these fields is an orange "CREATE ACCOUNT" button and a link "Already have an account? Login". The background of the page is dark with a faint image of a hospital room and some descriptive text about a prediction system.

Fig 4: Registration Page

6.1.4 Login Page: This is login page. In here user can login with their registered credentials.



The screenshot shows the same web browser at localhost:3000/#about. The navigation bar is identical. A central white modal titled "Login" contains the following fields: "Email Address" and "Password". Below these fields is an orange "LOG IN" button and a link "New user? Register | Forgot Password?". The background of the page is dark with a faint image of a hospital room and some descriptive text about a prediction system.

Fig 5: Login Page

6.1.5 Prediction Page: This is prediction page. In here, user can input their data and get prediction.

Welcome, 204g1a3204@srit.ac.in [LOGOUT](#)

New Sleep Analysis

Phone Number (Mandatory for Doctor Consultation)

Age Gender

Occupation

What do you do when you get angry?

Do you follow a healthy diet?

How do you usually handle a difficult situation?

Previous health issues?

Blood Pressure Heart Rate

Sleep Duration (hrs) Tea or Coffee (Cups/day)

BMI Category Snoring Frequency

Working Hours (Per Week)

[GENERATE PREDICTION](#)

Analysis History

Date	Result	Action
No records found.		

Fig 6: Prediction Page

6.1.6 Report Page: This is the report page. In here result will be display.

localhost:3000/view-report?id=1774150613330 [Summarize](#) [Update](#) [Chat](#)

AI HOSPITAL

AI-powered Sleep Analysis & Risk Assessment

AI Model: Ensemble ML (High Risk)

High Risk: Sleep Disorder Concern (ML Ensemble)

Risk Assessment Table:

Factor	Score	Weight	Contribution
Age	25	0.1	2.5
Gender	Male	0.1	1.0
Blood Pressure	120/80	0.2	24.0
Heart Rate	75	0.1	7.5
Sleep Duration	7 hrs	0.3	21.0
Tea or Coffee	2 cups	0.1	2.0
BMI Category	Underweight	0.2	4.0
Snoring Frequency	Never	0.1	1.0
Working Hours	9-5, 40 hrs	0.1	4.0
Total Risk Score			68.0

Why this is a High Risk Prediction:

- The patient's sleep duration is significantly below the recommended 7-9 hours.
- The patient's BMI is underweight, which is a known risk factor for sleep disorders.
- The patient's heart rate is slightly elevated, which may be related to sleep issues.
- The patient's working hours are long, which can lead to sleep deprivation.

Recommendations:

- Consult a sleep specialist for a thorough evaluation.
- Improve sleep hygiene: avoid caffeine, maintain a regular sleep schedule.
- Consider a sleep study to determine the exact cause of the sleep disorder.

Fig 7: Report Page

6.1.7 Admin Page: This is admin page, admin can view and manage system data and monitor the application.

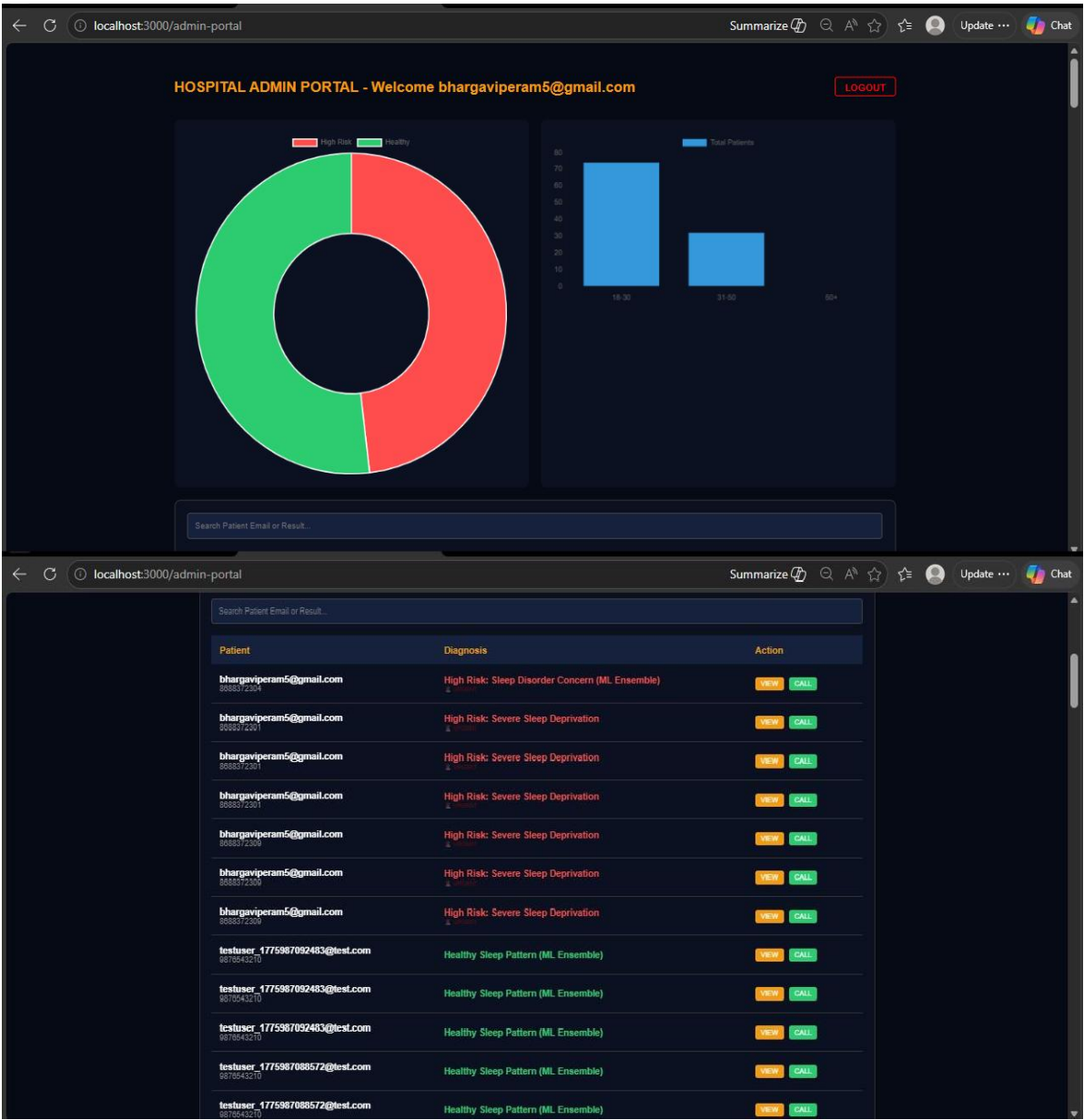


Fig 8: Admin page

7.Conclusion, Limitations and Scope for Future Work

7.1 Conclusion

This project develops an efficient system for predicting sleep disorders using an ensemble learning approach that combines multiple machine learning models such as KNN, SVM, and Random Forest. This project develops an efficient system for predicting sleep disorders using an ensemble learning approach that combines multiple machine learning models such as KNN, SVM and Random Forest. By applying Voting classifier the system improves prediction stability and overall reliability compared to individual models. It processes health and lifestyle data through data preprocessing, feature selection, model training, and evaluation to generate meaningful and accurate predictions. The system includes both user and system modules, where users can register, log in, enter patient data, and view results, while the system handles data processing, model execution, and result generation. This approach helps in early identification of sleep disorders, reduces dependency on costly diagnostic methods, and makes the solution more accessible and practical for real-world use. Additionally, the model is scalable and can be extended with larger datasets, more features, and advanced algorithms in the future, making it a strong foundation for improving healthcare decision-making and patient outcomes.

7.2 Limitations

While the Sleep Disorder Prediction System offers an efficient and user-friendly approach for early detection of sleep-related issues, it also has certain limitations. These limitations arise due to factors such as dependency on user-provided data, constraints of machine learning models, and the availability of quality datasets. Understanding these limitations is important to interpret the results correctly and to recognize that the system is intended to support, not replace, professional medical diagnosis.

7.3 Scope for Future Work

Future enhancements of this project can focus on improving the accuracy and usability of the sleep disorder prediction system by integrating advanced machine learning and deep learning models, including hybrid ensemble techniques. The system can be enhanced by incorporating additional real-time data sources such as wearable devices and sleep tracking

applications to provide more comprehensive and accurate inputs. Implementing real-time monitoring and prediction features would make the application more dynamic and useful for continuous health tracking, such as interactive dashboards and personalized recommendations, can enhance user experience. Applying model optimization techniques and expanding the system to detect multiple types of sleep disorders with multilingual support can increase its effectiveness and accessibility to a wider range of users.

8.References

- [1] Rahardian, F., & Rakasiwi, S. **“Interpretable Ensemble Models for Lifestyle-Based Sleep Disorder Prediction”** Journal of Applied Informatics and Computing, 2025.
- [2] Mukherjee, D., Dhar, K., Schwenker, F., & Sarkar, R. **“Ensemble of Deep Learning Models for Sleep Apnea Detection: An Experimental Study”** Sensors Journal, 2021.
- [3] Kumar, A., Bakshi, T., et al. **“Sleep Disorder Prediction Using Machine Learning”** IJLTEMAS, 2025.
- [4] Sarala, P., et al. **“Sleep Disorder Prediction using Advanced Machine Learning Techniques”** IJSAT, 2025.
- [5] Advanced Ensemble Model Study **“Advanced Sleep Disorder Detection using Multi-layered Ensemble Learning”** Frontiers in Artificial Intelligence, 2024.
- [6] Cheng, X., et al. **“SleepEGAN: GAN-Enhanced Ensemble Deep Learning Model for Sleep Stage Classification”** Biomedical Signal Processing, 2024.
- [7] Varghese, B., & Daniel, S. **“Sleep Disorder Prediction using Machine Learning”** International Journal of Science and Research, 2025.
- [8] Setiawati, M., et al. **“Prediction Analysis of Sleep Disorders Using Machine Learning-Based Techniques”** Jurnal Sistem Informasi Bisnis, 2025.
- [9] Alazaidah, R., et al. **“Potential of Machine Learning for Predicting Sleep Disorders: A Comprehensive Analysis”** Diagnostics Journal, 2023.
- [10] Seenivasan, S., et al. **“Sleep Disorder Classification Using AI: A Machine Learning Approach”** International Research Journal on Advanced Engineering Hub, 2025.
- [11] Yousefi, M. R., & Rahimi, R. **“Sleep Disorder Diagnosis Using EEG Signals and LSTM Deep Learning Method”** arXiv, 2025.
- [12] Ahadian, P., et al. **“Deep Time Series Analysis for Sleep Disorder Prediction using AI”** arXiv, 2024.
- [13] Alazaidah, R., Samara, G., et al. **“Potential of Machine Learning for Predicting Sleep Disorders: A Comprehensive Analysis of Regression and Classification Models”** Diagnostics Journal, 2024.
- [14] Lal, U., Vasanthseena, S. M., & Hoblidar, A. **“Temporal Feature Extraction and Machine Learning for Classification of Sleep Stages Using Polysomnography”** Brain Sciences, 2023.

- [15] Taspinar, Y. S., & Cinar, I. **“Prediction of Sleep Health Status, Visualization and Analysis of Data”** International Conference on Advanced Technologies, 2023.
- [16] Calderon, A., et al. **“Machine Learning and Bayesian Network Analysis for Insomnia Prediction”** BMC Psychiatry, 2024.
- [17] Yadav, P. K., et al. **“Sleep Disorder Detection Using Machine Learning Method”** IEEE Conference, 2023.
- [18] Liu, P., et al. **“Automatic Sleep Stage Classification Using Deep Learning”** Artificial Intelligence Review, 2024.
- [19] Zhang, H., et al. **“Simultaneous Sleep Stage and Sleep Disorder Detection from Multimodal Sensors Using Deep Learning”** Sensors Journal, 2023.
- [20] Scientific Reports Study **“Predicting Sleep Deprivation Using Machine Learning Algorithms”** Nature – Scientific Reports, 2024.